The Complete Structural Mereology of a Neurosymbolic Search-and-Rescue Manifold: A Unified Axiomatic System Specification

System Architecture Invariant Manifest

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1 Introduction and Architectural Core

We formalize the search-and-rescue plot for a central Fire-Station as an interlocking multi-graph Sequential Dynamic System (SDS) operating across distinct modal boundaries. Following Edward Zalta's Theory of Abstract Objects, the spatiotemporal execution plane maps to ordinary object exemplification (Fx), while the invariant control forests encode properties atemporally (xF). The ultimate objective of the system is the minimization of coordinate uncertainty to realize Mesarovic's Attainable Space (V'), culminating in physical vehicle extraction.

2 Axiomatic Metaphysics and the Causal Tracking Loop

The system trajectory of an asset is formalized as a higher-order relational tracking loop:

$$g(m, u, v) \equiv \exists y[R_{\text{system}}(m, u, y) \wedge E(m, u, y, v)] \tag{1}$$

where u represents environmental Navier-Stokes advection forces, m represents the external divine control input vector, y represents human or machine output behaviors, and E represents the absolute spiritual evaluation function.

3 The 30-Dimensional Integrated Attribute Matrix

To inject continuous meaning and speech-act context into the message-passing convolutions, the GNN ingests an integrated, multi-scale node attribute matrix $\boldsymbol{X} \in \mathbb{R}^{N \times 30}$. For each node i, the vector decomposes into six distinct socio-pragmatic and deontic tensor blocks:

$$\boldsymbol{x}_i = \left[\boldsymbol{e}_i^{\text{emp (16)}} \mid \boldsymbol{g}_i^{\text{spatial (2)}} \mid \boldsymbol{p}_i^{\text{ppl (3)}} \mid \boldsymbol{r}_i^{\text{rsa (3)}} \mid \boldsymbol{s}_i^{\text{searle (3)}} \mid \boldsymbol{t}_i^{\text{deontic (3)}} \right] \in \mathbb{R}^{30} \tag{2}$$

where $\boldsymbol{e}_i$ encapsulates the Barna loop splits and Pew macro-advection wavelets, $\boldsymbol{g}_i$ tracks continuous JAX coordinate steps, $\boldsymbol{p}_i$ and $\boldsymbol{r}_i$ pack the level-k PPL choice models and Rational Speech Act distributions, $\boldsymbol{s}_i$ maps the Searlean speech-act flags (X counts as Y in C), and $\boldsymbol{t}_i$ maintains the time-bounded legal compliance trajectories. The Graph Convolutional Network layer recursively convolutes these nodes over the empirical interaction topology:

$$h_i^{l+1} = \text{ReLU}\left(W^l \frac{1}{\sqrt{\deg_i}\sqrt{\deg_j}} \sum_{j \in \mathcal{N}_i} h_j^l \right) \tag{3}$$

4 Secure Distributed Multi-Agent System Initialization

We formalize a distributed multi-agent system composed of four decentralized nodes bounded by a security flow lattice and an epistemic authority partial order.

4.1 The Epistemic Authority Poset Lattice

Let $\mathcal{A} = \{a_0, a_1, a_2, a_3\}$ represent the set of decentralized neuro-symbolic agents. The agents are ordered over a strict bounded knowledge poset lattice governed by the partial ordering relation $\preceq$:

$$a_3 \preceq a_2 \preceq a_1 \preceq a_0 \tag{4}$$

where a_0 (`agent_0_master`) dominates the supreme coordinate position ($\top$) with an epistemic rank of 3, and a_3 (`agent_3_observer`) represents the public base rate layer ($\bot$) with a rank of 0. Parallel specialized knowledge bases map to comparable intermediate coordinates ($a_i \sim a_j$).

4.2 Localized Initialization Parameters

Each independent agent a_k is initialized as an isolated mereological tracking instance containing a personalized sub-vocabulary bounded by the global GNN mapping space, a discrete attention guardrail $\mathcal{K}_k$ written in Prolog, and an empirical decision forest $\mathcal{F}_k$ containing only local sensor-loop exemplars:

$$a_k \equiv \langle \mathcal{V}_k, \mathcal{K}_k, \mathcal{F}_k \rangle \tag{5}$$

Access control and transactional data flow between disparate security clearances (System Maintenance, Proprietary, Private, Public) are strictly monitored by Google Zanzibar-inspired causal ordering checks embedded directly inside the Model Context Protocol (MCP) logic layer, preventing unauthorized cross-talk or token leakage.